PRACTICE SET

Absolute Value Inequalities: Special Cases — Set 2

Name: _____ Date: _____

Directions: Isolate the absolute value first if needed. Then choose the type and write the solution in interval notation. Let c be the value that makes the inside equal 0.

A	No solution	$\varnothing$	e.g. $\lvert x-3\rvert \le -2$ or $\lvert x-3\rvert < 0$
B	All real numbers	$(-\infty, \infty)$	e.g. $\lvert x-3\rvert > -2$ or $\lvert x-3\rvert \ge 0$
C	Only one solution	$\{c\}$	e.g. $\lvert x-3\rvert \le 0 \Rightarrow \{3\}$
D	All reals except one number	$(-\infty, c) \cup (c, \infty)$	e.g. $\lvert x-3\rvert > 0 \Rightarrow (-\infty, 3) \cup (3, \infty)$

1. $|x+3| < -3$ ____ _____

2. $|2x-3| < 0$ ____ _____

3. $4\,|4x-2|+4 < 4$ ____ _____

4. $|2x+1| \ge 0$ ____ _____

5. $|2x+2| \le 0$ ____ _____

6. $2\,|5x-2|-2 < -14$ ____ _____

7. $|4x+4| > 0$ ____ _____

8. $|-x+3| > 0$ ____ _____

9. $3\,|3x+5|-1 \le -1$ ____ _____

10. $|3x-6| > 0$ ____ _____

11. $|2x+2| < -1$ ____ _____

12. $5\,|x+8|+3 > 3$ ____ _____

13. $|x+1| > -3$ ____ _____

14. $|2x+7| \ge -5$ ____ _____

15. $3\,|-3x-5|+7 > 7$ ____ _____

16. $|x+6| > 0$ ____ _____

17. $|x-9| > 0$ ____ _____

18. $5\,|x+3|+7 < 7$ ____ _____

19. $|-2x-5| \le -1$ ____ _____

20. $|5x-6| \ge -4$ ____ _____

21. $2\,|-2x-4|-6 \le -8$ ____ _____

22. $|5x+4| > -1$ ____ _____

23. $|3x-3| \ge -9$ ____ _____

24. $3\,|-3x-7|+6 \ge 6$ ____ _____

25. $|5x-8| \ge 0$ ____ _____

26. $|-x-1| \ge -7$ ____ _____

27. $5\,|5x-2|-3 \le -23$ ____ _____

28. $|x-2| \le 0$ ____ _____

29. $|5x+7| \le -4$ ____ _____

30. $3\,|x+9|-8 \le -8$ ____ _____

31. $|x+5| > 0$ ____ _____

32. $|x+2| \le 0$ ____ _____

33. $5\,|-x+8|+5 \le 5$ ____ _____

34. $|-2x+7| > -6$ ____ _____

35. $|2x-5| > 0$ ____ _____

36. $2\,|2x-3|+7 \ge 7$ ____ _____

37. $|5x+4| \le 0$ ____ _____

38. $|-x+9| < 0$ ____ _____

39. $3\,|x-1|+2 > 2$ ____ _____

40. $|-3x+7| \le 0$ ____ _____

41. $|-x-6| \le 0$ ____ _____

42. $4\,|-2x-1|+6 < -10$ ____ _____

43. $|-2x+6| \le 0$ ____ _____

44. $|x-8| > 0$ ____ _____

45. $5\,|-2x+9|+3 > -17$ ____ _____

46. $|3x-2| \le 0$ ____ _____

47. $|-2x-3| \le 0$ ____ _____

48. $4\,|2x-7|+1 > 1$ ____ _____

49. $|-3x-3| \le 0$ ____ _____

50. $|2x+1| > 0$ ____ _____

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